

FASTEST FINGER FIRST

For Four Players

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Fastest finger first technique is used in quiz shows (with rapid fire rounds) to find the order in which participants press the buzzer (to give an answer to the question). This project is designed for four players. For real-time applications, calling-bell switches can be used over tactile switches.

Circuit and working

Circuit diagram of the microcontroller (MCU)-based fastest finger first is shown in Fig. 1. It is built around 8-bit microcontroller AT89S51 (IC2), 230V AC primary to 9V, 300mA secondary step-down transformer (X1), 16x2 LCD (LCD1), 5V voltage regulator IC 7805 (IC1) and a few other components.

The power supply circuit is designed to provide +5V regulated DC voltage for the MCU and LCD1. Bridge rectifier (BR1) is used to convert 9V AC to 9V DC. IC1 regulates input DC

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - AT89S51, 8-bit MCU
- LED1 - 5mm LED
- BR1 - 1A bridge rectifier
- T1 - BC557 pnp transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3 - 1-kilo-ohm
- R2, R4-R7 - 10-kilo-ohm
- R8 - 100-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 100 μ F, 25V electrolytic
- C2 - 100 μ F, 16V electrolytic
- C3 - 10 μ F, 16V electrolytic
- C4-C5 - 33pF, ceramic disk
- C6 - 0.1 μ F, ceramic disk

Miscellaneous:

- LCD1 - 16X2 LCD
- S1-S5 - Tactile switch
- PZ1 - Piezo buzzer
- X1 - 230V AC, 50Hz primary to 9V, 300mA secondary transformer
- XTAL1 - 12MHz crystal oscillator
- 16-pin connector for LCD1

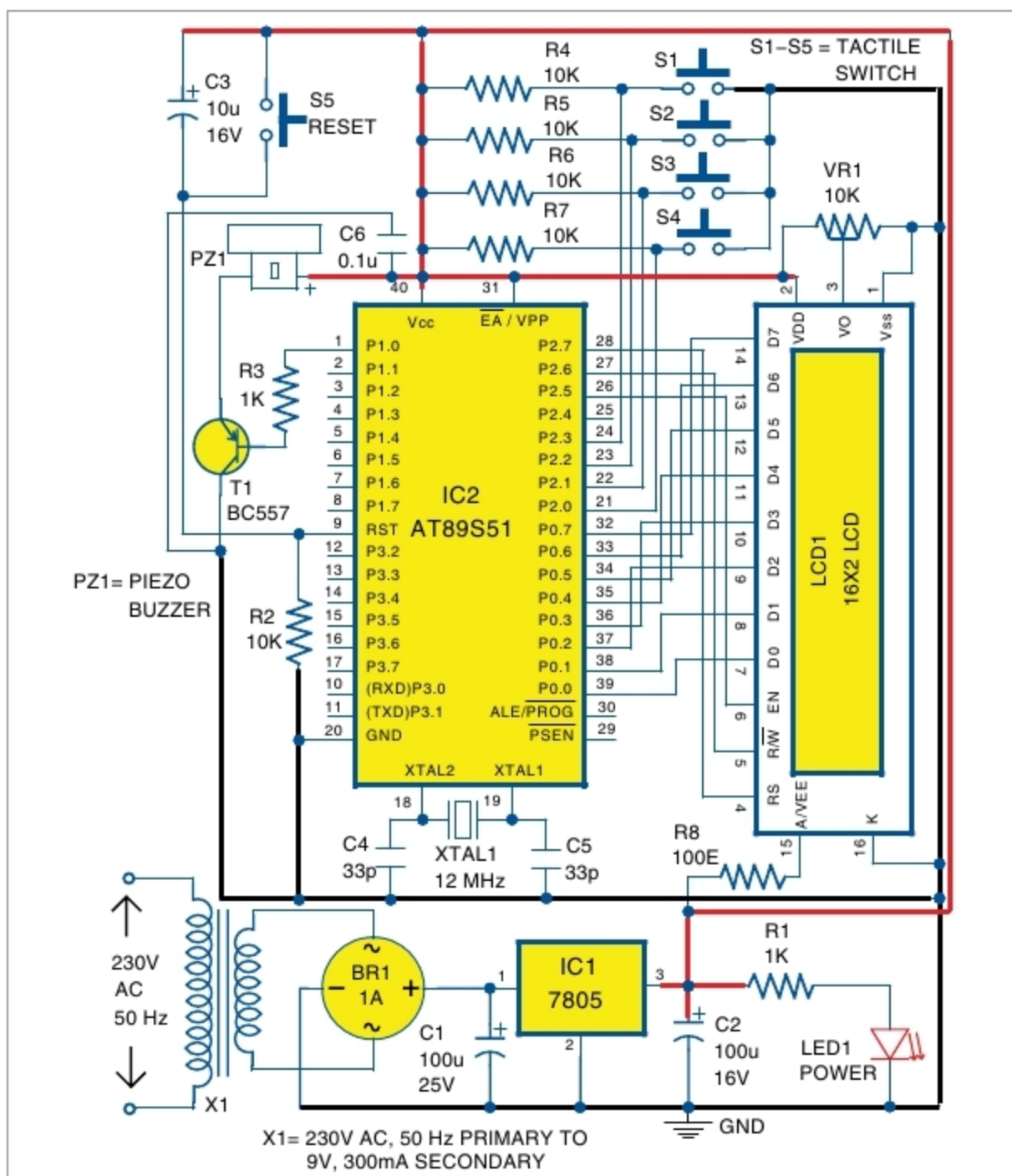


Fig. 1: Circuit diagram of fastest finger first

voltage to provide +5V DC. Capacitors C1 and C2 act as filters to remove noise from the power supply. LED1 is used as power supply indicator and resistor R1 limits the current flowing through LED1.

The oscillator section—comprising 12MHz crystal (XTAL1), capacitor C4 and C5—is used to provide clock pulse to the MCU. Resistor R2 and capacitor C3 form the power-on reset circuitry to reset the MCU whenever the supply is switched on. Switch S5 is used to

manually reset the MCU for fresh operation. Transistor T1 is used as the driver for piezo buzzer PZ1, and resistor R3 limits the base current of T1. PZ1 is used to provide a beep sound whenever a quiz cycle begins or ends. The buzzer is turned on when logic 0 is present in I/O pin P1.0 of IC2.

S1 through S4 are input switches for four players named A, B, C and D, respectively. Resistors R4 through R7 are used as pull-up resistors for switches S1 through S4, respectively.